IC101

Introductory Securities & Markets

Lecture 5 · Capital flows and the four sectors

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From one price to the whole economy

Four lectures built the pieces. Today we assemble them.

The journey so far

L1 Price, risk and return
L2 Balance sheets: debt and equity
L3 Banks: lending creates money
L4 Money and liquidity

Today is the final, biggest step.

Today: follow the money

That money we defined last week never sits still. It flows between four great sectors of the economy.

And here is the iron rule we will prove today:

one sector's deficit must be another's surplus. Always. It cannot be otherwise.

Bottom line: **today we follow money as it moves around the whole economy, and prove that every pound borrowed is a pound someone else saved.**

Today: the whole economy in four moves

One big idea: money flows, and the flows must balance

The four sectors Households, companies, government and the overseas world.

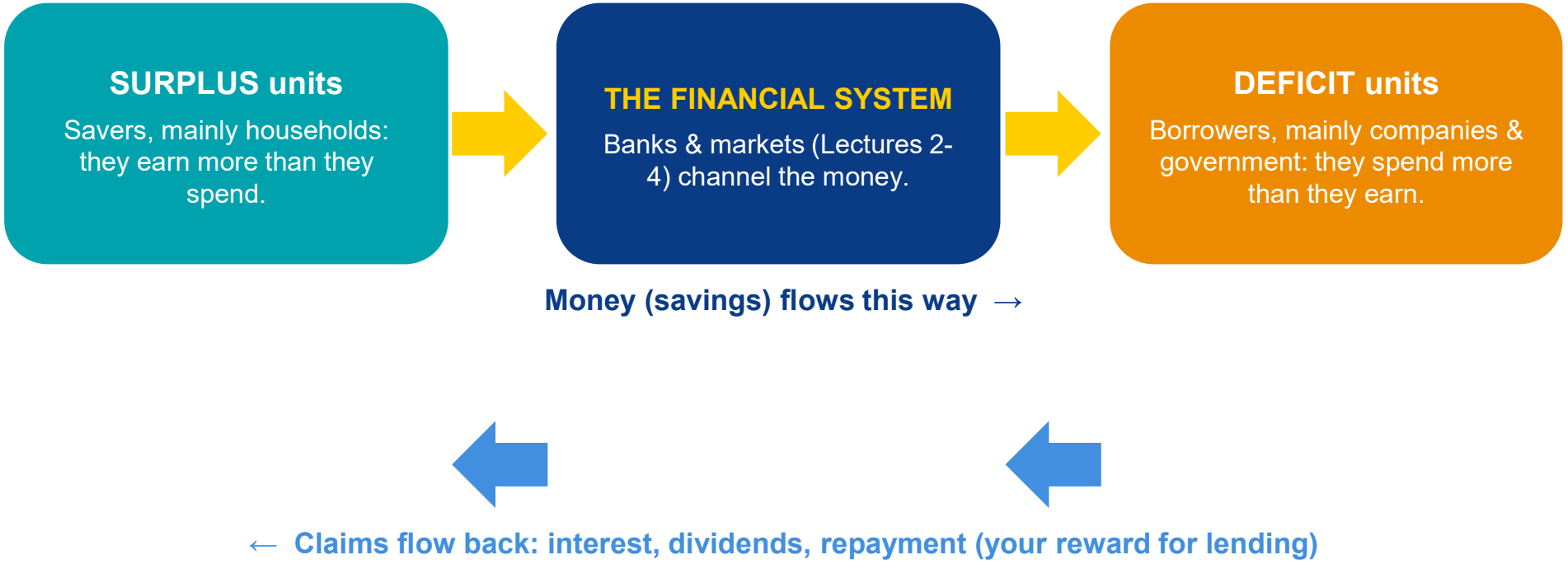
The flows between them Savers fund borrowers. Surpluses fund deficits, every time.

The balancing identity A simple equation that must always hold, tested on real UK data.

When the flows break Sovereign and pension crises, and why they hit so fast.

The economy is a giant circuit of money

Savers lend to borrowers, and claims flow back the other way



Bottom line: finance exists to move spare money from savers to borrowers, and send the reward (interest, dividends, repayment) back the other way. That is the whole game.

The four sectors of the economy

Every pound belongs to one of these four players



Households

Us. 'Natural persons'. We supply the labour and, in the end, own everything and bear all the risk.



Companies

Firms that produce goods and services. They invest, and usually need to borrow to do it.



Government

The 'sovereign': taxes, spends, and borrows. Near-infinite life, lowest domestic risk.



Overseas

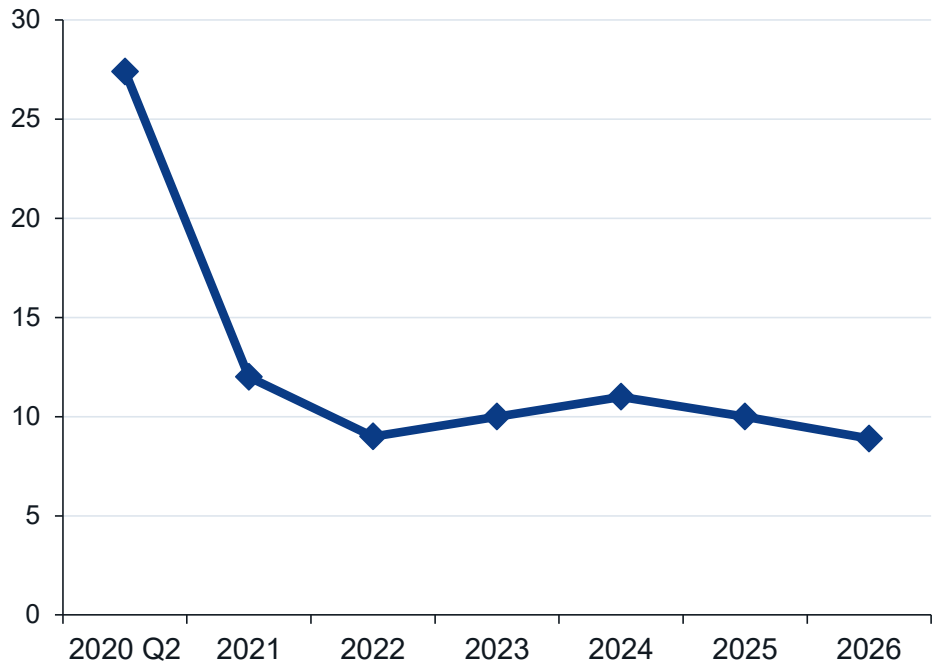
The rest of the world: every other country we trade and invest with.

Bottom line: the whole economy is just four players, you (households), firms, government and the rest of the world. Everything today is money moving between these four.

Households: the ultimate savers

The sector that funds everyone else, and carries the final risk

UK household saving ratio % of income saved



What this means for you

Households are the economy's net savers: together we earn more than we spend (about +2.3% of GDP).

Your saving matters even though the bank does not lend out your £100 (Lecture 3): deposits are what fund the bank's mortgages and business loans, so household saving ends up financing homes and factories.

We saved a record 27% of income in spring 2020, locked down with nothing to spend on, then just 8.9% by 2026 as bills bit.

And households bear ALL the risk: every loss, everywhere, lands on people in the end.

What is GDP? the total value of everything the UK produces in a year (about £2.8 trillion). So '+2.3% of GDP' just means 2.3% of the country's yearly output.

DEEPER READ **How and why to save: a free, plain-English guide (MoneyHelper)** ↗

UK households' saving ratio, ONS: record 27.4% in Q2 2020; 8.9% in Q1 2026. Household net lending about +2.3% of GDP, Q1 2026.

Companies: the investors who need capital

They turn savings into factories, products and jobs

Usually a net borrower

Companies exist to **INVEST**: build factories, develop products, hire people.

Investment usually costs more than a firm earns in cash right now, so the sector borrows the difference.

This is exactly why debt and equity (Lecture 2) and banks (Lecture 3) exist: to fund company investment.

Where your money goes to work

AstraZeneca borrows to build labs. Ørsted borrowed billions to build wind farms, until its debt was so high that in 2025 it had to ask shareholders for DKK 60bn (about £7bn) of new equity instead: Lecture 2's debt-or-equity choice.

A start-up raises equity to grow before it earns a penny of profit.

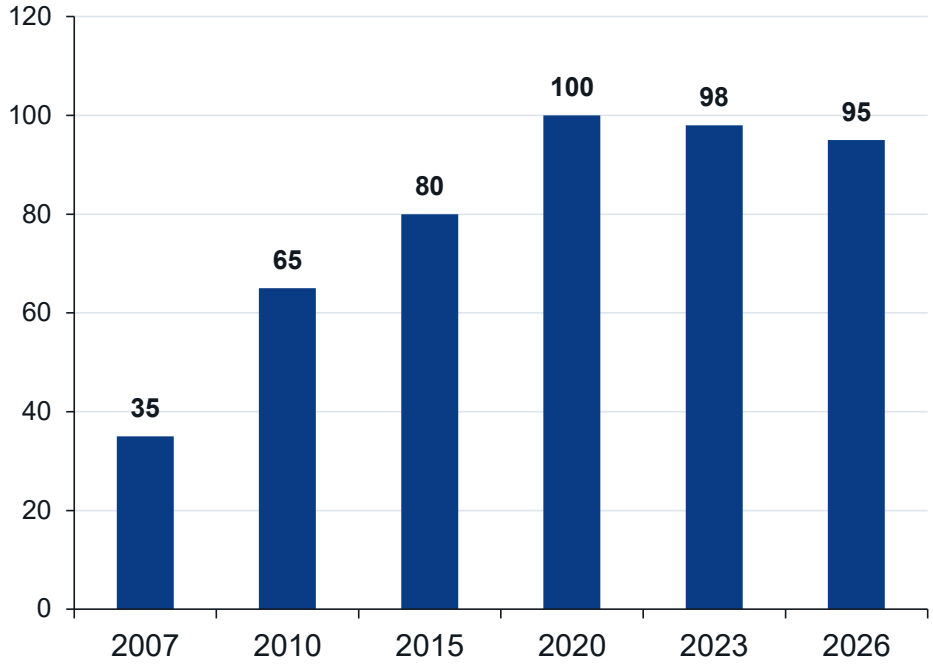
The household saving from the last slide is what ends up funding all of this.

Bottom line: companies borrow households' savings to invest. That is the whole reason banks and markets exist.

Government: the sovereign borrower

It spends more than it taxes, and borrows the rest from us

UK government debt % of GDP





A near-permanent deficit, and its price

The government usually spends more than it raises in tax. Each year's gap adds to the debt, now about 95% of GDP; it jumped after 2008 and 2020.

The price: interest on the debt cost £97.6bn in 2025-26, more than one and a half times the defence budget.

Lenders set the rate: in October 2026 the 30-year gilt yield passed 6%, the highest since 1998.

Still the safest UK borrower: it borrows in pounds and the Bank of England creates pounds, so it cannot run out.

The risk for lenders is being repaid in pounds worth less, so when they doubt the plan they demand more interest.

DEEPER READ What are government debt and debt interest? A plain guide (House of Commons Library) ↗

Net debt 94.9% of GDP (ONS). Interest £97.6bn, 2025-26 (ONS); defence ~£62bn. 30-yr gilt 6.04%, 7 Oct 2026.

Quick poll, grab your phones

Which sector INSIDE the UK is the biggest net saver?

- **A** The government
- **B** Companies
- **C** Households
- **D** Nobody, everyone borrows

Join at vevox.app

Overseas: the rest of the world

Money crosses borders, so the fourth sector is everyone else

What the 'current account' really is

Over a year, does the country SELL more to the world (exports) than it BUYS from it (imports)?

The UK buys **more than it sells**: we import more cars, phones, gas and food than we export. That gap is a deficit, about -2.8% of GDP.

To pay for buying more than we sell, we must borrow from foreigners, or sell them our assets.

Who owns Britain?

Roughly a third of UK government bonds are held by overseas investors.

Foreign firms own UK airports, football clubs, water companies and much of the FTSE.

'Kindness of strangers': a deficit country depends on foreigners keeping the faith.

Bottom line: a country that buys more from the world than it sells (like the UK) must borrow the difference from the world.

The clever twist: the money comes back

What happens when a country buys more than it sells depends on whose currency it owes



1 The UK: owes its own currency

Foreigners paid in pounds can only spend them here: on UK goods, gilts, firms or property. So the money always comes back.

If they want less of it: the pound falls and UK interest rates rise until they do. Painful, but never a dead stop.

2 The US and China: the flow is shrinking

China still sells the US far more than it buys. It used to lend the dollars straight back: its US government bonds peaked at $1.3tn in 2011.

By 2026 about $590bn. After Russia's reserves were frozen in 2022, China moved into gold and other assets. The dollars return through other buyers.

3 Sri Lanka: owes dollars

Its fuel, food and medicine are paid in dollars, which it cannot print. By 2022 tourism had collapsed and lenders stopped lending.

The money did not come back: reserves ran out, the rupee crashed, no fuel, and in April 2022 it defaulted. The imports simply stopped.

Bottom line: owe your own currency and the money comes back, at a price (a weaker currency, higher rates). Owe someone else's and it can simply stop.

The iron law: one deficit, someone's surplus

Every pound borrowed is a pound someone lent. Follow £100 of government borrowing

GOVERNMENT
spends £100 more than it taxes, so it borrows £100

→ pays it out as wages (say nurses: their income rises £100)

→ **Save £60: it buys gilts +£60**

Spend £40 on imports: lent back +£40

-£100 + £60 + £40 = £0 the government's £100 was lent by the people its spending paid

In textbook shorthand: (S - I) + (T - G) + (M - X) = 0

S - I the private sector
Saving minus Investment, by households and firms. Do they save more than they invest?
UK: +1.8% of GDP

T - G the government
Taxes minus Government spending. Does it tax more than it spends?
UK: -4.6% of GDP, a deficit

M - X overseas
Our iMports minus our eXports. If we buy more than we sell, foreigners lend it back.
UK: +2.8% of GDP

So what? A government deficit is always funded by someone else's surplus. If the government cuts its deficit, another balance must move: households save less, firms invest more, or we import less. UK: 1.8 - 4.6 + 2.8 = 0.

CLICK **Try the balancing sum with AI** ↗

Work the identity: find the missing balance

Give me any three sector balances and the fourth is fixed

S - I = households + companies; T - G = government; M - X = overseas. Know three, find the fourth:

Households	**+4**	given: households save more than they invest
Companies	**-1**	given: investing a little more than they save
Overseas	**+2**	given: we import more than we export; foreigners lend it back
Government	**?**	YOUR job: make all four add up to zero

4 - 1 + 2 = +5 of net lending, so the government must be -5: a deficit of 5% of GDP.

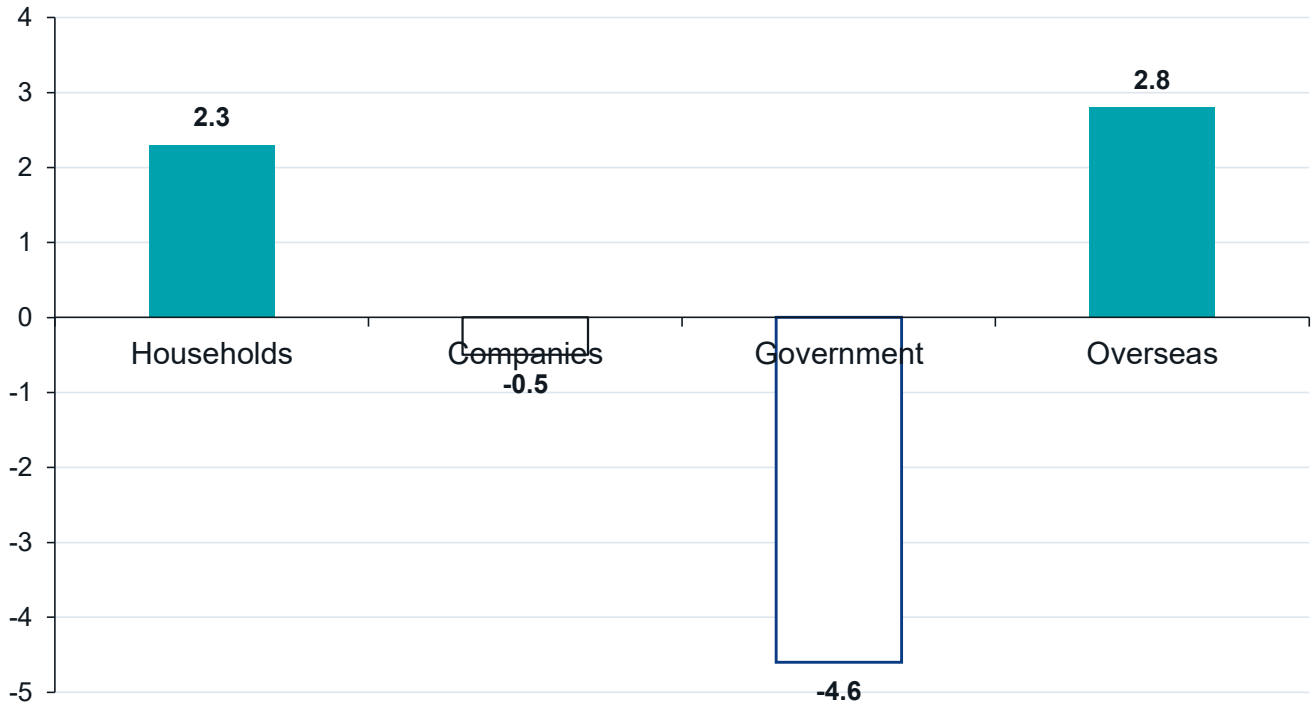
Now try: the government cuts its deficit to -3, and households and firms do not change. Overseas must fall to 0: we import less, so foreigners lend us less.

Bottom line: know three balances and the fourth is locked. Cut one sector's deficit and another balance must move to make room.

The UK today: it really does sum to zero

Households and foreigners are lending; the government is borrowing it, pound for pound

UK net lending by sector % of GDP, approx. 2026



Read it

Blue = surplus (a lender).

Black = deficit (a borrower).

Households (+2.3) and overseas (+2.8) fund the government's deficit (-4.6).

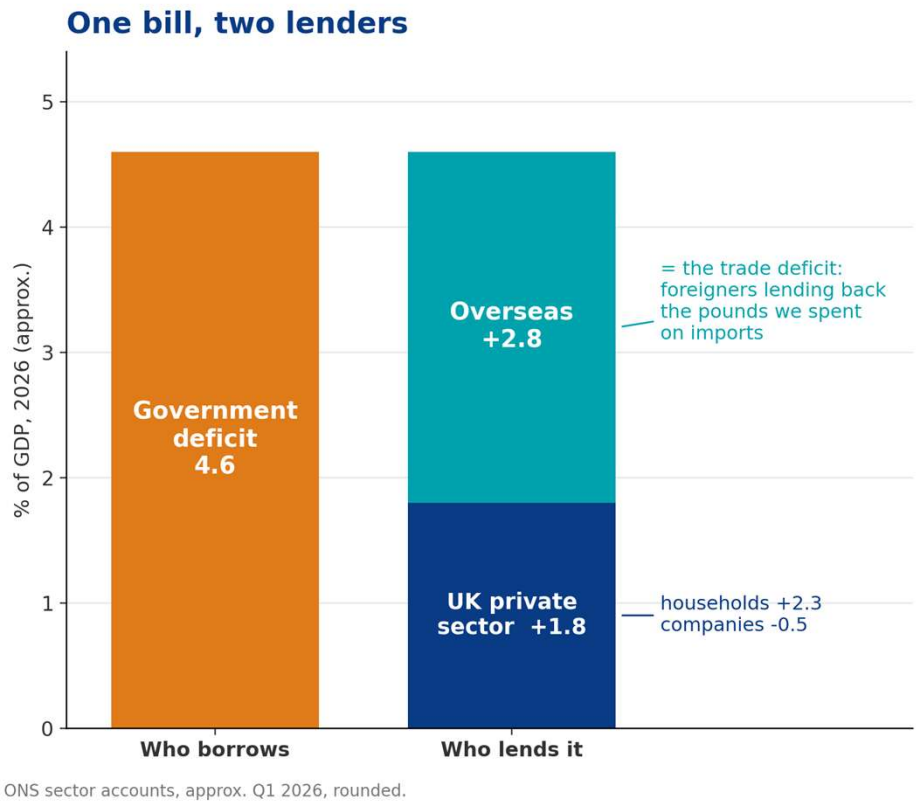
Add them: 2.3 - 0.5 - 4.6 + 2.8 = 0.

S - I = 2.3 - 0.5 = +1.8; T - G = -4.6; M - X = +2.8.

GDP = the UK's total yearly output (~£2.8tn). Net lending by sector, ONS UK sector accounts, approx. Q1 2026; rounded, sums to zero.

The UK's twin deficits

Two deficits, but only one bill: the trade deficit is one of the lenders



ONS sector accounts, approx. Q1 2026, rounded.

Twin deficits

Deficit 1, the government: spends 4.6% of GDP more than it taxes, and sells gilts to cover it.

Deficit 2, the country: imports 2.8% of GDP more than it exports. Those pounds come back as foreigners' lending: the overseas +2.8. It is not a second bill, it is one of the lenders.

So the 4.6 is funded by UK savers (+1.8 net) and foreigners (+2.8).

If the government cuts its deficit, another balance must move: households save less, firms invest more, or imports fall. That was the argument over UK austerity after 2010.

Bottom line: the UK borrows from its own savers and from abroad at the same time. Change one sector's balance and another must change too: the books always sum to zero.

How a government actually borrows: bonds

The deficit is funded by selling gilts to savers at home and abroad

1

Spend > tax

The government runs a deficit: it needs cash it does not have.

2

Issue gilts

It sells IOUs, UK government bonds called 'gilts', at auction.

3

Savers buy

Pension funds, banks, and overseas investors buy them with their surpluses.

4

Pay interest

The government pays interest for years, then repays. A claim flows back.

CLICK How does government borrowing work? Ask AI ↗

Bottom line: a government borrows by selling IOUs called gilts, and it is your pension fund and foreign savers who buy them. Gilts also set the 'risk-free' rate that prices every other loan and asset (Lecture 1).

When confidence snaps: the flows reverse

A sector that relied on others' money can be cut off overnight

Greece, 2010

Joined the euro in 2001 and borrowed cheaply for years. In October 2009 a new government admitted the deficit was over 12% of GDP, not under 4%.

Lenders panicked and Greek interest rates soared. In the euro, Greece could not print its own money or devalue: it could only cut, or be rescued.

Three bailouts (2010, 2012, 2015), a write-off for private lenders, and a quarter of the economy lost.

The UK, September 2022

23 September: £45bn of unfunded tax cuts with no official forecast. Investors sold gilts; 30-year yields jumped from under 4% to about 5% in three days.

Pension funds using 'LDI' had to post cash collateral as gilt prices fell, so they sold gilts, pushing prices lower still: the doom loop from Lecture 4.

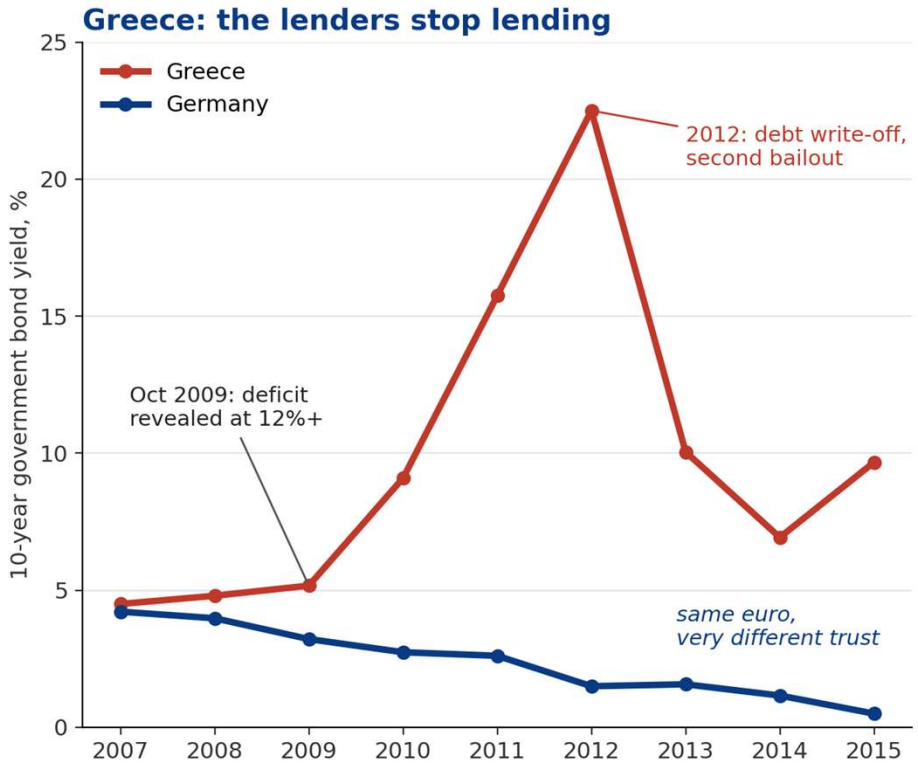
The Bank of England bought £19bn of gilts in two weeks. The Chancellor was sacked and Truss resigned within seven weeks.

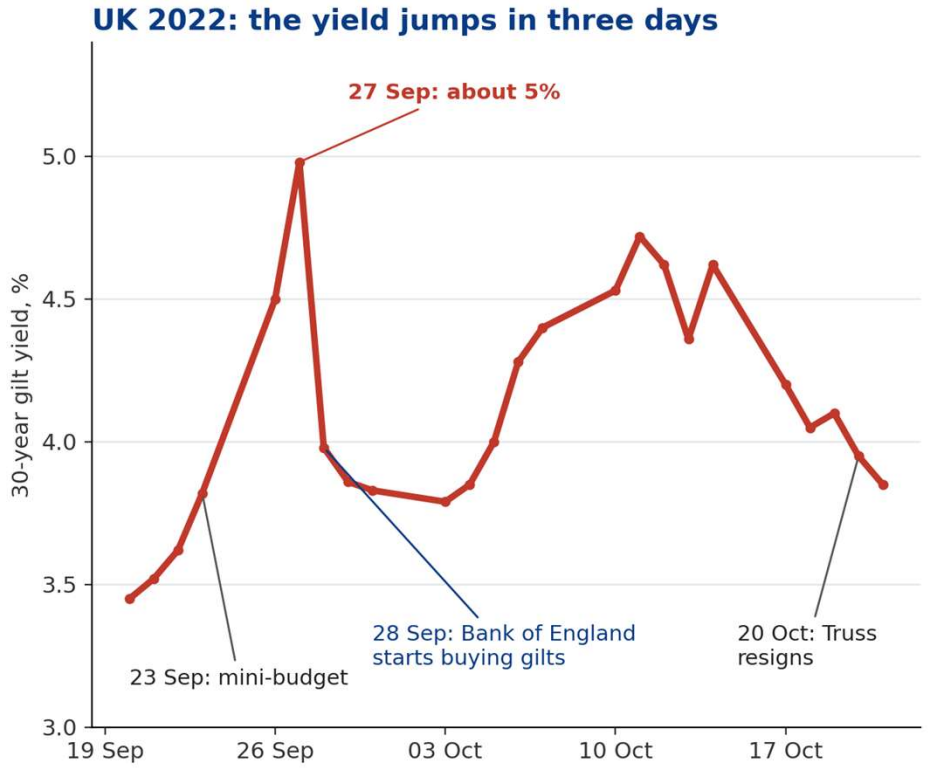
DEEPER READ How the 2022 mini-budget triggered a UK pension and gilt crisis (CNBC) ↗

CLICK What happened in the 2022 gilt crisis? Ask AI ↗

When confidence snaps, in numbers

Lenders do not stop gradually. They stop in months, or in days





Greece and Germany: annual average 10-year yields (Eurostat/ECB). UK: daily closing 30-year gilt yields, approximate.

Bottom line: **Greece could not print and was cut off; Zimbabwe printed and destroyed its money (Lecture 4). Either way, the lender's question is the same: will I be paid back in money worth having?**

Quick poll, grab your phones

In the 2022 UK crisis, WHY did selling gilts make things worse?

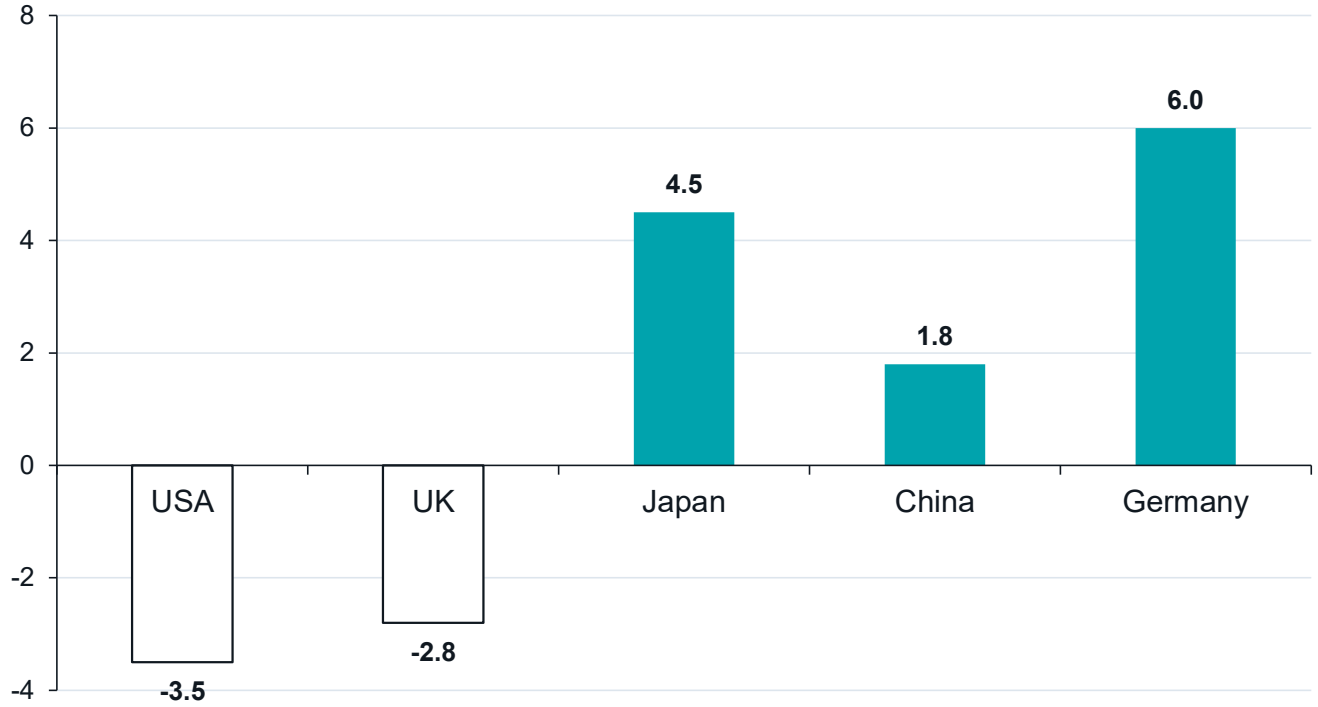
- **A** It had no effect on prices
- **B** Selling pushed prices down, forcing yet more selling
- **C** The government banned selling
- **D** Foreign investors bought them all instantly

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The world's savers and borrowers

Sell more to the world than you buy = you lend abroad. Buy more than you sell = you borrow abroad.

Do they sell more, or buy more, than the rest of the world? (current account, % of GDP)



What the bars mean

Sells more than it buys = surplus = lends the extra to the world.

Germany, Japan, China: factory nations that export more than they import.

Buys more than it sells = deficit = borrows from the world.

USA and UK: consume more than they make, so they borrow. The USA now owes the world about $26 trillion.

CLICK Why is the US the world's borrower? Ask AI ↗

DEEPER READ Which countries lend to the world, and which borrow? (IMF Blog) ↗

Current account balances, approximate recent figures, IMF. US net international investment position about -$26tn.

The global savings glut

Why some countries always lend, and others always borrow

The same rule runs the whole planet: some economies save far more than they invest, and lend the rest to the world.

The lenders (the 'glut')

China, Germany, Japan and the Gulf oil exporters run persistent surpluses, they save more than they spend and invest.

The borrowers

The United States above all, and the UK, run persistent deficits, funded by that wall of foreign saving.

Ben Bernanke called this the 'global savings glut'. It pushed world interest rates down and funded the deficit countries, and, many argue, helped inflate the 2008 housing bubble. The dollar's reserve-currency status lets the US borrow almost without limit, its 'exorbitant privilege'.

Bottom line: **At world scale too, one country's deficit is another's surplus. The glut of savings has to go somewhere, and it flows to the big borrowers.**

Why this is relevant to your career

The people who read these flows run the markets



Macro & rates

Hedge funds and banks bet on deficits, inflation and where interest rates go next.



Sovereign & policy

Treasuries, central banks and the IMF manage national balances and crises.



Global & EM investing

Investors move capital between countries, chasing returns and dodging sudden stops.



Bonds & FX

The gilt and currency desks price the risk-free rate that anchors all of finance.

Bottom line: every one of these jobs is, at heart, reading the four sectors and betting on the flows. Today you learned the map they use.

Five lectures, one system

From a single price to the flows of the whole world

L1	**Price, risk & return:** The atom: what a single asset is worth, and the risk-return trade-off.
L2	**Balance sheets:** The firm: how debt and equity fund it, and who bears the risk.
L3	**Banks:** The machine: lending creates money, and turns short-term deposits into long-term loans.
L4	**Money & liquidity:** The lifeblood: what money is (trust) and how fast it moves.
L5	**Capital flows:** The whole body: money circulating between four sectors, and the world.

Bottom line: It was one story all along: how society moves spare money to where it can do the most good, and shares the risk.

The highlights, know these cold



The four sectors: Households, companies, government, overseas. Households are the ultimate savers and risk-bearers.



Surpluses fund deficits: One sector's deficit is exactly another's surplus. The balances always sum to zero.



The identity: $(S - I) + (T - G) + (M - X) = 0$. In the UK, household saving and overseas lending fund the government's deficit.



Governments borrow via bonds: A deficit is financed by selling gilts; gilts set the risk-free rate for everything.



Flows run on confidence: When it snaps, reversals are fast and brutal: Greece 2010, the UK gilt crisis of 2022.

Exam check, can you answer these?

Same style as your mid-term test

Q. The UK government deficit is 4.6% of GDP and the trade deficit is 2.8%. What must the private sector's balance be?

A. +1.8% of GDP, a surplus. The balances sum to zero: private - 4.6 + 2.8 = 0. (If the trade deficit were the bigger one, the private sector would be in deficit too.)

Q. What is a 'gilt'?

A. A UK government bond: an IOU the government sells to fund its deficit.

Q. Why did selling gilts worsen the 2022 UK crisis?

A. It pushed prices down, triggering more forced selling: a liquidity spiral.

The foundations are complete

Lecture 5 of 5, and a handover

What you now own

You can read a price, a balance sheet, a bank, money itself, and the flows of a whole economy.

That is the complete foundation of finance. Genuinely, well done.

What comes next

Professor Brian Scott-Quinn takes over for the second half: sustainable finance and ESG.

You will now ask not just how capital flows, but WHERE it should flow: to a livable, sustainable future.

You have the machine. The second half asks the biggest question of all: what should we build with it?

Over to you

Before the second half

- Read Chapter 6 (the four sectors).
- Review Chapters 1-5 for the mid-term.
- Take the Blackboard quiz.

The one idea to keep

Surpluses fund deficits. Money flows between four sectors, and the balances always sum to zero.

Finance is how society moves spare money to where it does the most good.

YOUR SEMINAR THIS WEEK **Seminar 5: The four sectors & capital flows**

It goes deeper on today: the four sectors, sectoral balances, and how deficits are funded by other sectors.
To prepare: read Chapter 6 and come ready to work through the sectoral-balances identity.